

NVIDIA Nemotron-Cascade 2: IMO Gold Medal Reasoning with Only 3B Active Parameters

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30B MoE model with 3B active params achieves competition Gold Medals

3B Active Parameters activated per query out of 30B total (MoE)	20x Smaller than DeepSeek's Gold Medal model (671B)	3 Golds IMO 2025, IOI 2025, and ICPC World Finals
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NVIDIA released [Nemotron-Cascade 2](#), a 30 billion parameter mixture-of-experts model that activates only 3 billion parameters per query. It scored Gold Medal-level on the 2025 International Mathematical Olympiad (35 points), the International Olympiad in Informatics

(439.3), and the ICPC World Finals (10 of 12 problems). Only one other open-weight model has reached this tier: DeepSeek’s V3.2-Speciale, which uses 671B total parameters and 37B active. That makes Nemotron-Cascade 2 roughly 20x smaller. The [training recipe](#) combines Cascade RL, where reinforcement learning is applied in stages across math, code, and agentic tasks, with multi-domain on-policy distillation, which uses the best intermediate checkpoints to prevent skill regression during training. NVIDIA released the full model, SFT dataset, and RL dataset openly.

A model solving competition-level math with 3B active parameters changes serving economics. Previous models at this reasoning level required multi-GPU setups. Nemotron-Cascade 2 [runs on a single GPU](#) via vLLM with a 262K token context window. On coding, it scores 87.2 on LiveCodeBench v6 and 50.2 on SWE-bench Verified, competitive with much larger models. For teams that need strong reasoning but can’t afford to serve a 671B model, this is the new baseline.

The trend is consistent: training methods are compressing what used to require massive scale. [GLM-5](#) showed a 744B MoE matching proprietary models. Nemotron-Cascade 2 reaches similar reasoning capability at 30B. Better RL curricula and distillation strategies are proving worth more than an order of magnitude in parameters.

Sources:

- [Nemotron-Cascade 2 on HuggingFace](#)
- [Nemotron-Cascade 2 Paper \(arXiv\)](#)
- [Nemotron-Cascade 2 SFT Dataset](#)
- [Nemotron-Cascade 2 RL Dataset](#)

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